Department of Computer Science and Engineering
Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE 'B'
Course Code & Title: CS3491 Artificial Intelligence and Machine Learning
Name of the Faculty member (s): Dr.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit II / Bayesian Inference**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** 2b
- **Activity Chosen: Think Pair Share**
- **Justification:**
  Bayesian Inference involves probabilistic reasoning, prior knowledge, and updating beliefs — concepts that are conceptually rich and often challenging for students to internalize. Think-Pair-Share (TPS) is effective for teaching Bayesian Inference as it promotes deeper understanding of those complex concepts. TPS supports critical thinking, especially when tackling counterintuitive aspects of Bayesian logic. TPS transforms Bayesian inference into an engaging, collaborative learning process that enhances conceptual clarity and retention.
- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
  - The concept of Bayesian inference methods along with examples was discussed in a detailed manner. After that the Think- Pair- Share activity was conducted at the end of the class.
  - The instructor asked the students to think about the given question – "A factory wants to predict machine failures based on sensor data (temperature, vibration, past failures). How can Bayesian Inference help in predictive maintenance?"
  - Then the students were asked to think about this question in terms of Bayesian inference basics and how to apply this technique to find inference for the predictive maintenance.
  - After that they are asked to pair with their team member and discuss and sharing their responses among themselves with other partner and it is shown in Fig.1.
  - At the end, the students from the Team1 - Mr.Rishikesh & Mr.Lokesh Kannan, Team 2- Ms.Harini & Ms. Ashmitha, Team 3- Ms.Dharshini & Ms.Pushpakalai shared their insights to the entire class so that all of them get different perspective for finding the inference for the given question and it is shown in Fig 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	1	2	3	1	2	2	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO9**	**PO10**	**PSO3**
	3	**2**	**1**	**1**	**1**	**3**
Justification for correlation	The students apply the knowledge of probabilistic reasoning.	The students analyze and identify the inference methods	The students find the solution to the given question	The students work as a team to share their solution.	The students communicate the solution to others.	The concept of Bayesian inference is essential for handling uncertainty.

- **Images / Screenshot of the practice:**

Fig 1. Student Involvement

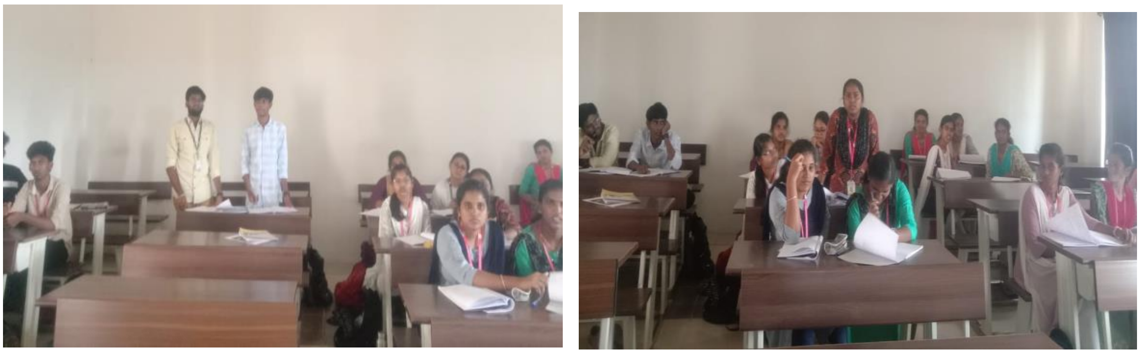

Fig 2. Share their views with classmates

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Students expressed that explaining concepts to their partner improved their clarity and retention.
    - Students felt that the interactive nature of the activity helped them break down the complex logic behind Bayesian Inference.

- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - This activity enhances understanding, deepens engagement, and builds confidence in tackling probabilistic concepts.
    - Encourages collaboration and communication among students.

- ***Challenges faced in implementation:***
    - Some of the students may struggle to find the answer using the probability concepts.
    - Ensuring all pairs stay on topic and engage meaningfully.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2023-2024/IP/CS3391_GSP_23_24_TPS.pdf
- https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share